

Automation Hydroponic Plant Protection from rain

1. Introduction/Problem Statement:

"Hydroponics planting method that utilizes water containing nutrients as a growing medium offers advantages such as not requiring large land areas and easy maintenance (Ilhamdi et al., 2018). The hydroponic method allows vegetable cultivation to be carried out in urban areas such as apartment balconies, house terraces, and even indoors for those with very limited space (Harsono, 2020). Despite these advantages, planting on balconies or terraces does not protect plants from exposure to rain. Rain can cause overwatering and nutrient leaching, which may hinder optimal nutrient absorption and excessive nutrient addition. Furthermore, the impact of rain can lead to unsatisfactory harvests or even complete crop failure. While indoor planting offers protection from rain, it requires higher investment costs for ultraviolet lighting as an artificial sun. The use of a mini greenhouse outdoors, although providing protection against weather conditions, greenhouses still require human intervention to adjust temperature and humidity (Kusumawati, Untoro, 2022). This project aims to protect hydroponic plants from rain by designing an automatic cover that will shield plants with plastic when it rains. This will ensure that hydroponic plants can grow optimally despite rainy weather. The project will focus on designing a purely analog circuit with actuators and sensors to achieve the desired functionality."

2. Feasibility & Scope:

"This project can be completed within the timeframe given in this course. The required electronic components, such as rain drop sensor, DPDT relay, limit switch, transistor, resistor, diode, and capacitor, along with actuator components like a gearbox DC motor, are available and affordable. Materials for the plant cover, and mica plastic are also accessible. Our level of knowledge of basic circuit analysis and analog circuit design to manage this project. The project will focus on designing and implementing a core circuit that detects rain. The circuit will use a rain drop sensor for water droplets, and transistors as switches to trigger the relay. This focused scope allows concentration on essential principles of an automatic plant cover and ensures the project is manageable. More complex designs with additional features can be considered for future projects."

3. Feedback Integration:

"We have discussed our project idea with the lab assistant. He suggested that using a power-adaptor would be beneficial. This would eliminate the need to change the batteries every time it runs out of power. We agree that this is a valuable addition. I will incorporate a power-adaptor into my circuit design. He also cautioned me about the risk of water that can damage the circuit. He suggested that we should make a case to protect the circuit from contact with water."